3-3. Equation 3-3 suggests how a velocity selector for particles or mixtures of different particles all having the same charge can be made. Suppose you wish to make a velocity selector that allows undeflected passage for electrons whose kinetic energy is 5.0×10^4 eV. The electric field available to you is 2.0×10^5 V/m. What magnetic field will be needed?

3-4. A cosmic ray proton approaches Earth vertically at the equator, where the horizontal component of Earth's magnetic field is 3.5×10^{-5} T. If the proton is moving at 3.0×10^6 m/s, what is the ratio of the magnetic force to the gravitational force on the proton?

3-5. An electron of kinetic energy 45 keV moves in a circular orbit perpendicular to a magnetic field of 0.325 T. (*a*) Compute the radius of the orbit. (*b*) Find the period and frequency of the motion.

3-6. If electrons have kinetic energy of 2000 eV, find (*a*) their speed, (*b*) the time needed to traverse a distance of 5 cm between plates D and E in Figure 3-1, and (*c*) the vertical component of their velocity after passing between the plates if the electric field is 3.33×10^3 V/m.

3-7. In J. J. Thomson's first method (see Problem 3-44), the heat capacity of the beam stopper was about 5×10^{-3} cal/°C and the temperature increase was about 2°C. How many 2000-eV electrons struck the beam stopper?

3-8. On drop #16, Millikan measured the following total charges, among others, at different times:

25.41×10^{-19} C	17.47×10^{-19} C	12.70×10^{-19} C
20.64×10^{-19} C	19.06×10^{-19} C	14.29×10^{-19} C

What value of the fundamental quantized charge e do these numbers imply?

3-9. Show that the electric field needed to make the rise time of the oil drop equal to its field-free fall time is $\mathscr{E} = 2mg/q$.

3-10. One variation of the Millikan oil drop apparatus arranges the electric field horizontally, rather than vertically, giving charged droplets acceleration in the horizontal direction. The result is that the droplet falls in a straight line that makes an angle θ with the vertical. Show that

$$\sin\theta = q\mathscr{E}/bv_t'$$

where v_t' is the terminal speed along the angled path.

3-11. A charged oil droplet falls 5.0 mm in 20.0 s at terminal speed in the absence of an electric field. The specific gravity of air is 1.35×10^{-3} and that of oil is 0.75. The viscosity of air is 1.80×10^{-5} N·s/m². (*a*) What are the mass and radius of the drop? (*b*) If the droplet carries two units of electric charge and is in an electric field of 2.5×10^5 V/m, what is the ratio of the electric force to the gravitational force on the droplet?

Section 3-2 Blackbody Radiation

3-12. Find λ_m for blackbody radiation at (*a*) $T = 3$ K, (*b*) $T = 300$ K, and (*c*) $T = 3000$ K.

3-13. Use the result of Example 3-4 and Equations 3-4 and 3-6 to express Stefan's constant in terms of h, c, and k. Using the known values of these constants, calculate Stefan's constant.

3-14. Show that Planck's law, Equation 3-18, expressed in terms of the frequency f, is

$$u(f) = \frac{8\pi f^2}{c^3}\frac{hf}{e^{hf/kT} - 1}$$

3-15. As noted in the chapter, the cosmic microwave background radiation fits the Planck equation for a blackbody at 2.7 K. (*a*) What is the wavelength at the maximum intensity of the spectrum of the background radiation? (*b*) What is the frequency of the radiation at the maximum? (*c*) What is the total power incident on Earth from the background radiation?

3-16. Find the temperature of a blackbody if its spectrum has its peak at (*a*) $\lambda_m = 700$ nm (visible), (*b*) $\lambda_m = 3$ cm (microwave region), and (*c*) $\lambda_m = 3$ m (FM radio waves).

3-17. If the absolute temperature of a blackbody is doubled, by what factor is the total emitted power increased?

3-18. Calculate the average energy $\overline{E}$ per mode of oscillation for (*a*) a long wavelength $\lambda = 10\,hc/kT$, (*b*) a short wavelength $\lambda = 0.1\,hc/kT$, and compare your results with the classical prediction kT (see Equation 3-9). (The classical value comes from the equipartition theorem discussed in Chapter 8.)

3-19. A particular radiating cavity has the maximum of its spectra distribution of radiated power at a wavelength of 27.0 μm (in the infrared region of the spectrum). The temperature is then changed so that the total power radiated by the cavity doubles. (*a*) Compute the new temperature. (*b*) At what wavelength does the new spectral distribution have its maximum value?

3-20. A certain very bright star has an effective surface temperature of 20,000 K. (*a*) Assuming that it radiates as a blackbody, what is the wavelength at which $u(\lambda)$ is maximum? (*b*) In what part of the electromagnetic spectrum does the maximum lie?

3-21. The energy reaching Earth from the Sun at the top of the atmosphere is 1.36×10^3 W/m², called the *solar constant*. Assuming that Earth radiates like a blackbody at uniform temperature, what do you conclude is the equilibrium temperature of Earth?

3-22. A 40-W incandescent bulb radiates from a tungsten filament operating at 3300 K. Assuming that the bulb radiates like a blackbody, (*a*) what are the frequency f_m and the wavelength λ_m at the maximum of the spectral distribution? (*b*) If f_m is a good approximation of the average frequency of the photons emitted by the bulb, about how many photons is the bulb radiating per second? (*c*) If you are looking at the bulb from 5 m away, how many photons enter your eye per second? (The diameter of your pupil is about 5.0 mm.)

3-23. Use Planck's law, Equation 3-18, to derive the constant in Wein's law, Equation 3-5.

Section 3-3 The Photoelectric Effect

3-24. The wavelengths of visible light range from about 380 nm to about 750 nm. (*a*) What is the range of photon energies (in eV) in visible light? (*b*) A typical FM radio station's broadcast frequency is about 100 MHz. What is the energy of an FM photon of the frequency?

3-25. The orbiting space shuttle moves around Earth well above 99 percent of the atmosphere, yet it still accumulates an electric charge on its skin due, in part, to the loss of electrons caused by the photoelectric effect with sunlight. Suppose the skin of the shuttle is coated with Ni, which has a relatively large work function $\phi = 4.87$ eV at the temperatures encountered in orbit. (*a*) What is the maximum wavelength in the solar spectrum that can result in the emission of photoelectrons from the shuttle's skin? (*b*) What is the maximum fraction of the total power falling on the shuttle that could potentially produce photoelectrons?

3-26. The work function for cesium is 1.9 eV, the lowest of any metal. (*a*) Find the threshold frequency and wavelength for the photoelectric effect. Find the stopping potential if the wavelength of the incident light is (*b*) 300 nm and (*c*) 400 nm.

3-27. (*a*) If 5 percent of the power of a 100-W bulb is radiated in the visible spectrum, how many visible photons are radiated per second? (*b*) If the bulb is a point source radiating equally in all directions, what is the flux of photons (number per unit time per unit area) at a distance of 2 m?

3-28. The work function of molybdenum is 4.22 eV. (*a*) What is the threshold frequency for the photoelectric effect in molybdenum? (*b*) Will yellow light of wavelength 560 nm cause ejection of photoelectrons from molybdenum? Prove your answer.

3-29. The NaCl molecule has a bond energy of 4.26 eV; that is, this energy must be supplied in order to dissociate the molecule into neutral Na and Cl atoms (see Chapter 9). (*a*) What are the minimum frequency and maximum wavelength of the photon necessary to dissociate the molecule? (*b*) In what part of the electromagnetic spectrum is this photon?

3-30. A photoelectric experiment with cesium yields stopping potentials for $\lambda = 435.8$ nm and $\lambda = 546.1$ nm to be 0.95 V and 0.38 V, respectively. Using these data only, find the threshold frequency and work function for cesium and the value of h.

3-31. Under optimum conditions, the eye will perceive a flash if about 60 photons arrive at the cornea. How much energy is this in joules if the wavelength of the light is 550 nm?

3-32. The longest wavelength of light that will cause emission of electrons from cesium is 653 nm. (*a*) Compute the work function for cesium. (*b*) If light of 300 nm (ultraviolet) were to shine on cesium, what would be the energy of the ejected electrons?

Section 3-4 X Rays and the Compton Effect

3-33. Use Compton's equation (Equation 3-25) to compute the value of $\Delta\lambda$ in Figure 3-17*d*. To what percent shift in the wavelength does this correspond?

3-34. X-ray tubes currently used by dentists often have accelerating voltages of 80 kV. What is the minimum wavelength of the x rays they produce?

3-35. Find the momentum of a photon in eV/c and in kg · m/s if the wavelength is (*a*) 400 nm, (*b*) 1 Å = 0.1 nm, (*c*) 3 cm, and (*d*) 2 nm.

3-36. Gamma rays emitted by radioactive nuclei also exhibit measurable Compton scattering. Suppose a 0.511-MeV photon from a positron-electron annihilation scatters at 110° from a free electron. What are the energies of the scattered photon and the recoiling electron? Relative to the initial direction of the 0.511-MeV photon, what is the direction of the recoiling electron's velocity vector?

3-37. The wavelength of Compton-scattered photons is measured at $\theta = 90°$. If $\Delta\lambda/\lambda$ is to be 1 percent, what should the wavelength of the incident photon be?

3-38. Compton used photons of wavelength 0.0711 nm. (*a*) What is the energy of these photons? (*b*) What is the wavelength of the photons scattered at $\theta = 180°$? (*c*) What is the energy of the photons scattered at $\theta = 180°$? (*d*) What is the recoil energy of the electrons if $\theta = 180°$?

3-39. Compute $\Delta\lambda$ for photons scattered at 120° from (*a*) free protons, (*b*) free electrons, and (*c*) N_2 molecules in air.

3-40. Compton's equation (Equation 3-25) indicates that a graph of λ_2 versus $(1 - \cos\theta)$ should be a straight line whose slope h/mc allows a determination of h. Given that the wavelength of λ_1 in Figure 3-17 is 0.0711 nm, compute λ_2 for each scattering angle in the figure and graph the results versus $(1 - \cos\theta)$. What is the slope of the line?

3-41. (*a*) Compute the Compton wavelength of an electron and a proton. (*b*) What is the energy of a photon whose wavelength is equal to the Compton wavelength of (1) the electron and (2) the proton?

Level II

3-42. When light of wavelength 450 nm is incident on potassium, photoelectrons with stopping potential of 0.52 V are emitted. If the wavelength of the incident light is changed to 300 nm, the stopping potential is 1.90 V. Using *only* these numbers together with the values of the speed of light and the electron charge, (*a*) find the work function of potassium and (*b*) compute a value for Planck's constant.

3-43. Assuming that the difference between Thomson's calculated e/m in his second experiment (Figure 3-19) and the currently accepted value was due entirely to his neglecting the horizontal component of Earth's magnetic field outside the deflection plates, what value for that component does the difference imply? (Thomson's data: $B = 5.5 \times 10^{-4}$ T, $\mathscr{E} = 1.5 \times 10^4$ V/m, $x_1 = 5$ cm, $y_2/x_2 = 8/110$.)

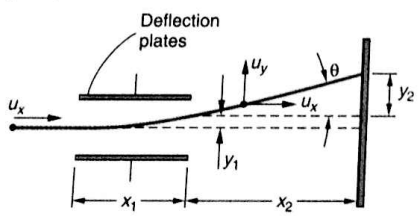

Figure 3-19 Deflection of the electron beam in Thomson's apparatus. The deflection plates are D and E in Figure 3-1. Deflection is shown with magnetic field off and the top plate positive. The magnetic field is applied perpendicular to the plane of the diagram and directed into the page.

3-44. In his first e/m experiment, Thomson determined the speed of electrons accelerated through a potential ΔV by collecting them in an insulated beam stopper and measuring both the total collected charge Q and the temperature rise ΔT of the beam stopper. (*a*) Show that with those measurements he could obtain an expression for e/m in terms of the speed of the electrons and the directly measured quantities. (*b*) Show that the expression obtained in (*a*) together with the result of Problem 3-2 enabled Thomson to compute e/m in terms of directly measured quantities.

3-45. Data for stopping potential versus wavelength for the photoelectric effect using sodium are

λ, nm	200	300	400	500	600
V_0, V	4.20	2.06	1.05	0.41	0.03

Plot these data in such a way as to be able to obtain (*a*) the work function, (*b*) the threshold frequency, and (*c*) the ratio h/e.

3-46. Prove that the photoelectric effect cannot occur with a completely free electron, i.e., one not bound to an atom. (*Hint:* Consider the reference frame in which the total momentum if the electron and the incident photon is zero.)

3-47. When a beam of monochromatic x rays is incident on a particular NaCl crystal, Bragg reflection in the first order (i.e., with $m = 1$) occurs at $\theta = 20°$. The value of $d = 0.28$ nm. What is the minimum voltage at which the x-ray tube can be operating?

3-48. A 100-W beam of light is shone onto a blackbody of mass 2×10^{-3} kg for 10^4 s. The blackbody is initially at rest in a frictionless space. (*a*) Compute the total energy and momentum absorbed by the blackbody from the light beam, (*b*) calculate the blackbody's velocity at the end of the period of illumination, and (*c*) compute the final kinetic energy of the blackbody. Why is the latter less than the total energy of the absorbed photons?

3-49. Show that the maximum kinetic energy E_k, called the Compton edge, that a recoiling electron can carry away from a Compton scattering event is given by

$$E_k = \frac{hf}{1 + mc^2/2hf} = \frac{2E_\gamma^2}{2E_\gamma + mc^2}$$

3-50. The x-ray spectrometer on board a satellite measures the wavelength at the maximum intensity emitted by a particular star to be $\lambda_m = 82.8$ nm. Assuming that the star radiates like a blackbody, (*a*) compute the star's surface temperature. (*b*) What is the ratio of the intensity radiated at $\lambda = 70$ nm and at $\lambda = 100$ nm to that radiated at λ_m.

3-51. Determine the fraction of the energy radiated by the Sun in the visible region of the spectrum (350 nm to 700 nm). Assume that the Sun's surface temperature is 5800 K.

3-52. Millikan's data for the photoelectric effect in lithium are shown in the table:

Incident λ (nm)	253.5	312.5	365.0	404.7	433.9
Stopping voltage V_0 (V)	2.57	1.67	1.09	0.73	0.55

(*a*) Graph the data and determine the work function for lithium. (*b*) Find the value of Planck's constant directly from the graph in (*a*) . (*c*) The work function for lead is 4.14 eV. Which, if any, of the wavelengths in the table would not cause emission of photoelectrons from lead?

Level III

3-53. This problem is to derive the Wien displacement law, Equation 3-5. (*a*) Show that the energy density distribution function can be written $u = C\lambda^{-5}(e^{a/\lambda} - 1)^{-1}$, where C is a constant and $a = hc/kT$. (*b*) Show that the value of λ for which $du/d\lambda = 0$ satisfies the equation $5\lambda(1 - e^{-a/\lambda}) = a$. (*c*) This equation can be solved with a calculator by the trial-and-error method. Try $\lambda = \alpha a$ for various values of α until λ/a is determined to four significant figures. (*d*) Show that your solution in (*c*) implies $\lambda_m T$ = constant and calculate the value of the constant.

3-54. This problem is one of *estimating* the time lag (expected classically, but not observed) for the photoelectric effect. Assume that a point light source emits 1 W = 1 J/s of light energy. (*a*) Assuming uniform radiation in all directions, find the light intensity in eV/s · m^2 at a distance of 1 m from the light source. (*b*) Assuming some reasonable size for an atom, find the energy per unit time incident on the atom for this intensity. (*c*) If the work function is 2 eV, how long does it take for this much energy to be absorbed, assuming that all of the energy hitting the atom is absorbed?